



Gradeable Work Types — Research Digest

Dark visual edition • 2026-09-24

Card t_a2d9bdc1 • GRADE • Research digest (no app implementation)

Chuck Broaddus / Kelyra • Confidential

Source: dated research digests under notes/company • graphics illustrate locked recommendations.

Executive recommendation

Locked recommendation

Hybrid: canonical work_kind vocabulary (seedable, not locked) + teacher/school syllabus categories with weights; map work_kind → default category at create; score schemes per column. v1 expands seed + schemes + capture hints; defer full weighted engine / SBG.

Decision cards

Model

Hybrid vocab+cats

Math

Syllabus weights

Schemes

Multi score types

v1

Seed + hints

Hybrid: work_kind vocab + syllabus categories

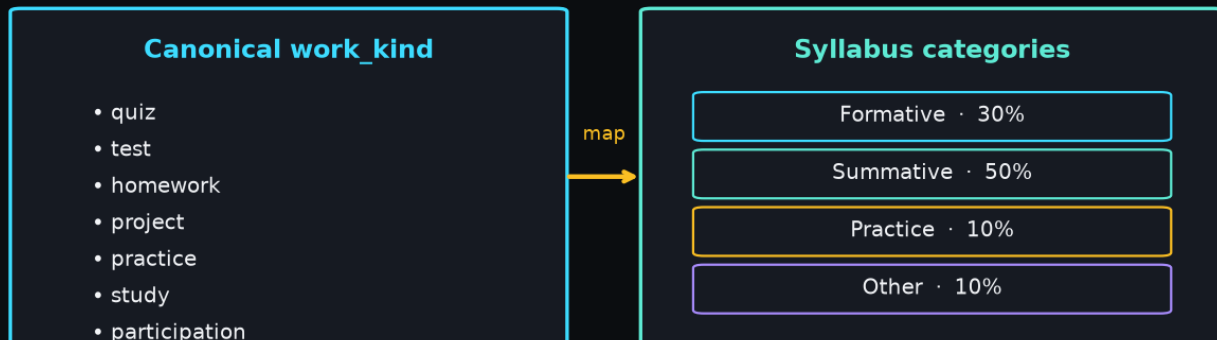


Figure: work_kind ↔ syllabus categories.

Gradeable Work Types — Research + Plan Digest

Date: 2026-09-24 Author: Chief of Staff / Grok Bot (Kelyra) Status: Ready for Chuck review SoT pair: [notes/company/gradeable-work-types-research.md](#) · [notes/company/gradeable-work-types-plan.md](#) Card: [t_a2d9bdc1](#) Access date for cited public URLs: 2026-09-24 (America/Chicago)

This dated file is a readable pack for Chuck. Full taxonomy and sources live in the SoT pair above.

Top recommendations

1. Hybrid model (not fixed enum, not pure free-text): canonical [work_kind](#) vocabulary for AI/filters/presets plus teacher-/school-named syllabus categories for grade math (AVG).
2. v1 Spring Baptist = Phase A labels: expand [work_kind](#) seeds + picker + capture type hint + Christian extras ([memorization](#) / Bible-friendly); add [complete_incomplete](#) score scheme; keep Approve gate.
3. Do not ship weighted finals in this card's Eng follow-on — that is AVG ClassSyllabus; be honest in trial that averages stay simple until AVG.
4. Copy SIS/LMS pattern: categories first, assignments belong to one category, optional category weights, per-column include-in-final (PowerSchool, Canvas groups, Google Classroom, Skyward, FACTS).
5. Skip for v1: SIS of record, auto-publish AI grades, quiz Author player, SBG library, Dojo-style behavior points, office-locked catalogs (unless Chuck picks locked).
6. Default participation / behavior / citizenship out of academic average ([include_in_average=false](#)); policy can re-enable later.
7. AI can draft photo homework / quizzes / essays / labs; manual for live participation, PE demos, conduct — still Approve for anything that becomes a grade.

Research (full SoT)

Gradeable Work Types Research (GWT-R1)

Date: 2026-09-24 Author: Chief of Staff / Grok Bot (Kelyra) Status: Ready for Chuck review Revision: R1 Cards: [t_a2d9bdc1](#) Access date for all URLs below: 2026-09-24 (America/Chicago)

Related (do not rebuild):

- [research/05-gradebooks-skill-plans-incentives.md](#) — AI scoring, standards trends, ILP rarity
- [notes/company/avg-research-syllabus-conventions.md](#) (AVG-R1) — district weights, floors, drop/replace
- [notes/company/avg-research-rubrics-vs-syllabus.md](#) (AVG-R2) — syllabus ≠ rubric ≠ category
- [notes/company/avg-spec-syllabus-ia.md](#) (AVG-P1) — ClassSyllabus + SyllabusCategory sketch
- [notes/authoring/kinds-metrics.md](#) — Author emits [kind=lesson](#); quiz/test are category labels
- [docs/mvp.md](#) / [docs/data-model.md](#) / [docs/vision.md](#) — simple grade book; Approve gate
- Plan SoT: [notes/company/gradeable-work-types-plan.md](#)

Constraints for this card: Research + recommendation only. No app code, no SQL apply, no Hermes AI staffing, no force-push.

Executive summary

Teachers grade far more than “homework / quiz / test.” A practical taxonomy for Kelyra spans academic artifacts, process/soft measures, specials & elementary checklists, non-assignment marks, and calculation mechanics (weights, drop-lowest, retakes, incomplete). Every major SIS/LMS (PowerSchool, Skyward, Canvas, Google

Classroom, FACTS/RenWeb) converges on the same shape: teacher- or school-defined categories + assignment columns that belong to one category + optional category weighting — not a fixed global enum of work kinds.

Verdict — hybrid model (recommended):

1. Canonical work-kind vocabulary (enum-like keys for AI capture hints, filters, calendar chips, Author labels) — seedable, not locked.
2. Class syllabus categories (teacher-/school-named buckets with weights) — AVG-P1 already sketched; this is the grade-math surface.
3. Map work-kind → default category at create time; teacher can refile.
4. Score schemes per column: numeric / pass-fail / letter / rubric-level / complete-incomplete / narrative-only.
5. v1 Spring Baptist trial: expand category seed + score schemes + capture type hint; keep simple grade book law (Approve before grade); defer full weighted engine / SBG / behavior as GPA until AVG syllabus ships.
6. Do not become district SIS of record; do not auto-publish AI scores.

1. Problem framing for Kelyra

Reality	Product implication
Capture → Needs → Approve / KEYGRADE already files homework photos and practice into ...	Need richer labels so teachers can support “anything they might want” without inventing a second gradeb...
Live <code>assignments.category</code> defaults ...	Seed list is thin vs classroom reality; no class-level weight contract yet
MVP M9: “No weights, no categories, no CSV yet” (<code>docs/mvp.md</code>)	Research must separate taxonomy coverage from weighted average engine
Vision: not official school of record; no SIS passback in v1	Flexibility for private/Christian trial (Spring Baptist / FACTS-like expectations) without PowerSchool parity
Author: quiz/test are category labels, not players (<code>kinds-metrics...</code>)	Work-type research must not spawn <code>kind=quiz</code> HTML runtimes

CEO ask (2026-09-24): exhaustive practical taxonomy; SIS/LMS patterns; modeling recommendation; needed vs desired for Spring Baptist v1 vs later; gaps vs current product; open questions.

2. Exhaustive taxonomy (practical)

Organized as facets a teacher may combine on one grade-book column. One assignment row is usually one primary work kind + one syllabus category + one score scheme + term + include_in_average.

2.1 Academic artifacts (product of student work)

Work kind (canonical key)	Typical labels teachers use	Notes
<code>homework</code>	Homework, HW, home practice	Classic nightly; often low weight / formative
<code>classwork</code>	Classwork, seatwork, in-class	Same day; may be completion-based
<code>warmup</code>	Warm-up, bell ringer, do-now	High frequency, low stakes
<code>exit_ticket</code>	Exit ticket, exit slip	Capture-friendly; short formative
<code>quiz</code>	Quiz, check, pop quiz	Short assessment; may be hidden until publish (calendar law)
<code>test</code>	Test, chapter test, unit test	Summative mid-length
<code>exam</code>	Exam, midterm, final, semester exam	High weight; often own category
<code>project</code>	Project, PBL, performance task	Multi-day; often rubric
<code>lab</code>	Lab, lab report, science investigation	Specials + secondary science

Work kind (canonical key)	Typical labels teachers use	Notes
essay	Essay, paper, DBQ, CER, constructed response	AI rubric drafts (research/05)
presentation	Presentation, speech, oral	Partial AI; often rubric + live observe
portfolio	Portfolio, writing folder, binder check	Cumulative; may be checklist
discussion	Discussion, Socratic, seminar	Participation-adjacent; Canvas treats Discussions as graded items
practice	Assigned practice, intervention set	Kelyra closed loop — keep distinct from original capture
worksheet	Worksheet, packet	Often homework/classwork alias
notebook	Notebook check, interactive NB	Completeness + quality mix
reading_log	Reading log, minutes	Elementary + ELA
memorization	Memory verse, poem, fact fluency	Christian / classical school common (FACTS examples)

Score schemes per column

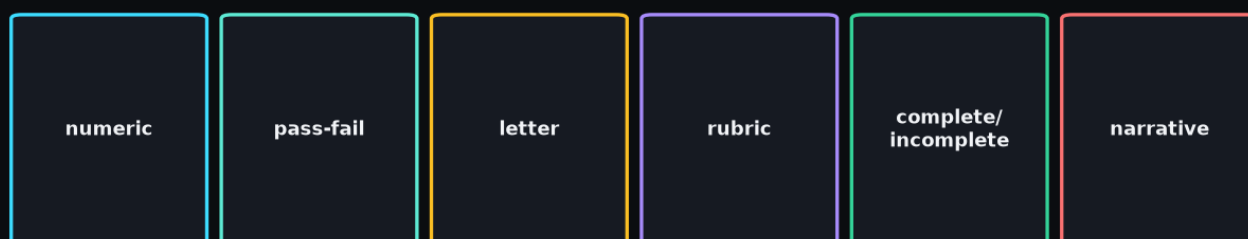


Figure: Score schemes per column.

2.2 Process / soft / non-content measures

Work kind	Typical labels	Gradebook caution
participation	Participation, engagement	Equitable policies often ban or shrink (AVG-R1 Springfield)
effort	Effort, work habits	Often separate from academic average
behavior	Behavior, conduct	Frequently not in academic average; office discipline separate
citizenship	Citizenship, stewardship	Private/Christian report cards
collaboration	Group work / teamwork	Rubric criterion more often than category
preparedness	Materials, homework turned in	Overlaps attendance/missing
organization	Binder, locker, desk	Elementary checklist

2.3 Attendance-linked & administrative marks (often not “assignments”)

Kind	How schools handle	Kelyra note
attendance_grade	Some still grade attendance	Prefer out of academic average; Ride/attendance elsewhere
missing / incomplete / ...	Score codes, not work types	FACTS grading codes pattern; cell state not category
late	Policy flag + optional penalty	Syllabus rule, not a work kind
extra_credit	Separate category or 0-max assignment	Equitable guidelines discourage; support as optional category weight 0 or excluded
retake / correction	Replaces or averages prior	Policy on category (AVG-P1), not a permanent work kind — may be second submission / linked column
drop_lowest	Category rule	Calc engine, not taxonomy row

2.4 Formative vs summative (meta-facet, not exclusive kind)

Band	Examples	Typical weight role
Formative	warm-up, exit ticket, homework, classwork, quiz (sometimes)	0–40% in equitable policies
Summative	test, exam, project, major essay, portfolio defense	60–100%
Ungraded formative	practice drafts, diagnostics	include_in_average=false

Kelyra already has `weight_band = none \ | daily \ | major \ | custom` — useful hint, not a substitute for syllabus category weights (AVG-P1).

2.5 Score schemes (how the cell is recorded)

Scheme	Use	Averages?
numeric (points or 0–100 %)	Default secondary	Yes
letter	A–F / ESNU	Map via scale for average, or display-only
pass_fail	Labs safety, some quizzes	Live <code>score_scheme</code> ; never average as % without map
complete_incomplete	Warm-ups, reading logs	Completion % or exclude
rubric_level	1–4 / Emerging–Mastered	Standards-based path; average later
checklist	Kindergarten skills, IEP progress (careful FERPA)	Often narrative + marks
narrative	Report-card comments, Glow/Grow	Not a number; <code>teacher_note</code> / <code>parent_sentence</code>
either	Live Kelyra: numeric or pass_fail	Keep

2.6 Specials, elementary, secondary, private/Christian nuances

Context	Common gradeables	Modeling note
Elementary	Checklists, ESNU, reading levels, behavior/citizenship separate	Favor complete/incomplete + narrative; light weights
Kindergarten	Developmental checklists, not % averages	Do not force 100-pt average

Context	Common gradeables	Modeling note
Specials (PE / music / art / library)	Performance, participation, skill demo, concert	Often fewer categories; performance + participation
Secondary	Weighted categories, exams, labs, essays	Full syllabus weights
AP / Dual / IB	External policy; may differ weights	Per-class flag (AVG-R1); not GPA engine in v1
Private / Christian (FACTS)	Memory verse, chapel participation, stewardship, Bible quiz	Seed Christian-school presets; teacher-named categories
Standards-based (SBG)	Standards as “columns,” attempts over time (Otus pattern in research/05)	Later; keep attempts + trend, not only last color

2.7 Non-assignment “grades” teachers still want on a report

- Term comment / narrative report-card blurb
- Conduct / citizenship mark separate from academic average
- Effort mark
- Standards mastery summary (later)
- Attendance summary (join Ride / office — not invent gradebook attendance)
- Mid-quarter progress flag

These are report artifacts, not always [assignments](#) rows. Model as syllabus-adjacent fields or term marks later — do not overload every comment into a weighted category.

3. SIS / LMS patterns (what to copy vs keep flexible)

3.1 Pattern catalog

Pattern	Who	Copy?	Keep flexible?
A. Categories first, then assignments	PowerSchool, Skyward, FACTS, Canvas groups	Yes — cannot create scored columns without a bucket	Allow school seed + teacher rename
B. Category weighting (weights sum ~100%)	PowerSchool Category Weighting; Classroom Weighted; Canvas Assignment Groups; Skyward percent-to-categories; F...	Yes for AVG — not required for GWT taxonomy alone	Allow Total Points mode later
C. Total points	PowerSchool default; Classroom; Canvas unweighted	Later option	v1 can stay “simple cells” until syllabus
D. Term / reporting-period weighting	PowerSchool Term Weighting	Later (quarters → semester)	
E. Drop highest/lowest in group	Canvas assignment group rules	Desired with AVG	
F. Count in final override per assignment	PowerSchool “Count in Traditional Final Grade”	Maps to include_in_average	Keep teacher override
G. Extra points / extra credit	PowerSchool Extra Points; Skyward 0-max or No Count	Optional	Do not encourage as default policy
H. Standards on assignment	PowerSchool standards count; Skyward standards GB; Otus attempts	Later SBG	Capture skill_gaps already exist
I. Score codes (M, I, Excused, Absent)	FACTS grading codes; most SIS	Needed for missing workflow	Cell state ≠ work kind
J. Admin-locked vs teacher-editable categories	Skyward often admin; PowerSchool district may lock; Classroom teacher-owned	Private school: teacher/office seed	Spring Baptist: start teacher-editable

3.2 Vendor signals (public docs; accessed 2026-09-24 CT)

Vendor	Categories	Calc modes	Source
PowerSchool / PowerTeacher Pro	Homework/tests-style categories; category must exist before assignment	Total points, term weighting, standards weighting, category weighting, mixed	Grade Calculation Types; Working with Assignments
Canvas	Assignment Groups (teacher-named); Discussions/Quizzes are still "assignments"	Weight final grade based on groups; drop lowest/highest	Yale Canvas weighted groups; CU Boulder Assignment Groups
Google Classroom	Grade categories (Homework, Essays, Quizzes, Tests...)	No overall / Total points / Weighted by category (weights sum 100%)	Set up grading; Add grade category
Skyward	Categories often admin-defined; color-coded	Total points vs percents assigned to categories; weight multiplier per assignment	District guides e.g. Dover SD Skyward guide; SchoolDay Skyward Gradebook Guide
FACTS / RenWeb	Categories required before assignments; weights; term checkboxes; color	Points vs Percent; ParentsWeb shows categories	Create Assignment Categories in FACTS SIS (PDF); RenWeb University / school training manuals

Takeaway to copy: flexible named categories + optional weights + per-column include flag. Takeaway to skip for v1: mixed double-count formulas, full standards-weighting into traditional grade, district-locked category catalogs, SIS grade passback.

3.3 Alignment with research/05

- Teacher-in-the-loop AI scoring (CoGrader, Formative Luna, Gemini Classroom) attaches to artifacts (essay, free response) — work kinds that are photographable or typed.



Figure: Capture → kind → category map.

- Standards trend graphs (Otus) need attempts over time, not only category averages.
- Participation/behavior are weak incentive levers vs parent messaging — do not center product thesis on behavior points (Dojo-style).

4. Gaps vs current Kelyra product

Area	Live / documented today	Gap
Capture kinds	homework \	voice_note
Assignment category	Free text; default homework; short seed list in data-model	Not exhaustive; no school presets; no Christian seeds
Syllabus / weights	weight_band, weight_percent...	AVG HOLD; weighted final not first-class
Score schemes	numeric \	pass_fail \
Practice loop	Distinct practice assignment from capture	Keep; ensure category defaults (e.g. formative / practice)
Author	kind=lesson only; category labels quiz/test	Correct — do not invent quiz player
Grade book scope	Simple teacher grid; not of record	Still true; taxonomy must not imply SIS replacement
Behavior / citizenship	Listed in category examples	No separate conduct report mark
Missing/incomplete codes	Not a first-class cell vocabulary in docs skimmed	Needed for teacher trust
Specials / K checklists	Not modeled	Later templates
SBG / standards library	Explicitly out of MVP (L9)	Later; skill_gaps are teacher-named

5. Modeling recommendation (Kelyra)

5.1 Hybrid (not pure enum, not pure free-text)

School (optional seed catalog) ■■■ ClassSyllabus (AVG) ■■■ SyllabusCategory { key, name, weight_pct, rules.. } ■■■ Assignment { work_kind, category_key, score_scheme, term, include_in_average, ... } ■■■ Submission { approved_score | mark_code | narrative }

Layer	What it is	Why
work_kind	Canonical vocabulary (enum-like string keys from §2)	AI capture hint, filters, calendar chips, Author label mapping, analytics
category_key	Points at syllabus category (teacher-named)	Grade math — Homework 10% / Tests 40%
score_scheme	How the cell is entered	Numeric vs checklist vs narrative
mark_code (later)	M / I / Exc / Abs / Late	Cell state independent of kind

Defaulting rule: creating an assignment with `work_kind=quiz` suggests category “Quizzes” if present, else creates/asks. Teacher always wins.

Do not force 1:1 work_kind ↔ category. Teachers often put quizzes and tests in one “Assessments” bucket, or split “Major” / “Daily.”

5.2 Fixed enum vs teacher-defined vs hybrid

Option	Pros	Cons	Verdict
Fixed global enum only	Simple AI	Teachers invent names; Christian schools blocked	Reject
Pure free-text only	Flexible	No filters, bad AI, broken weights	Reject as sole model
Hybrid	Presets + rename + custom	Slight complexity	Recommend

5.3 What capture / AI can grade vs manual-only

Work kinds	AI draft score/gaps?	Manual-only?
Photographed homework, worksheets, exit tickets, quizzes (paper), essays, labs (written)	Yes — existing analyze path; teacher Approves	
Digital lesson / practice sets	Yes — metrics → draft; Approve	
Live presentations, PE skill demo, oral	Partial (rubric assist later)	Observation scores often manual
Participation, behavior, citizenship, effort	No reliable AI	Manual (and often out of average)
Memorization (verse)	Possible with audio later	v1 manual or typed check
Narrative comments	AI draft one-liner already (parent_sentence)	Teacher edits

Law unchanged: nothing is a grade until Approve ([docs/data-model.md](#), vision).

5.4 Relationship to AVG syllabus epic

This GWT card feeds AVG: expands the seed list of category/work kinds and clarifies facets. It does not replace AVG-P1 ClassSyllabus tables. Weighted calculation remains AVG's job.

6. Needed vs desired (Spring Baptist v1 vs later)

Needed for trial / near-term (taxonomy + labeling — not full SIS)

1. Expand canonical `work_kind` seed + UI picker (homework, classwork, warm-up, exit ticket, quiz, test, exam, project, lab, essay, presentation, practice, participation, other + Christian extras: memorization / Bible).
2. Map `work_kind` → default category label on create; allow override.
3. Extend `score_scheme` toward complete/incomplete (+ keep numeric / `pass_fail`).
4. Capture flow: optional type hint (exit ticket vs homework) so Needs inbox and grade column are labeled correctly — still one photo → Approve.
5. Preserve Approve gate; AI drafts only.
6. Document that weights remain simple / AVG HOLD unless syllabus ships before trial — do not pretend weighted finals exist.

Desired later

1. Full ClassSyllabus weighted engine (AVG-P1).
2. Drop-lowest / retake-cap / late policy knobs.
3. Mark codes (missing/incomplete/excused).
4. Letter scales + ESNU elementary presets.
5. `Rubric_level` / analytic rubrics per assignment.
6. Standards-based attempts + trends (research/05 Otus pattern).
7. Conduct/citizenship separate from academic average.
8. Specials & kindergarten checklist templates.

v1 Spring Baptist trial — ship vs defer



Figure: Ship vs defer for Spring Baptist.

9. School-admin locked category catalogs (Skyward-like) if multi-teacher office demands.

10. SIS/LMS category sync / passback (explicitly later / out of MVP).

Out of scope

- Replacing FACTS/PowerSchool as official grade book of record
- Auto-publishing AI grades
- ClassDojo-style behavior point economy as core thesis
- Full state standards library (MVP L9)

7. Open questions for Chuck

1. For Spring Baptist, prefer FACTS-like teacher-built categories or a school-seeded locked list office controls?
2. Should participation / citizenship / behavior ever sit inside academic average, or always separate marks?
3. Is memory verse / Bible quiz a first-class preset for Christian templates, or generic “memorization”?
4. Trial priority: richer labels only first, or accelerate AVG weighted syllabus before Spring Baptist?
5. Kindergarten / early elementary on the trial roster — need checklist/ESNU in v1 or numeric-only OK?
6. Should pop quizzes default to calendar-hidden until Publish (existing calendar law) regardless of work_kind?

Sources

- [S1] PowerSchool — Grade Calculation Types — <https://ps.powerschool-docs.com/powerteacher-pro/latest/grade-calculation-types>
- [S2] PowerSchool — Working with Assignments — <https://ps.powerschool-docs.com/powerteacher-pro/latest/working-with-assignments>
- [S3] Google Classroom — Set up grading — <https://support.google.com/edu/classroom/answer/9184995>
- [S4] Google Classroom — Add a grade category — <https://support.google.com/edu/classroom/answer/9194393>
- [S5] Canvas @ Yale — Creating Weighted Groups — <https://help.canvas.yale.edu/a/914596-creating-weighted-assignments-groups>
- [S6] CU Boulder — Assignment Groups — <https://oit.colorado.edu/services/teaching-learning-applications/canvas/help/instructor-support/assignment-groups>
- [S7] FACTS — Create Assignment Categories (Catholic Schools BQ PDF) — <https://catholicschoolsbq.org/wp-content/uploads/2025/09/Create-Assignment-Categories-in-FACTS-SIS.pdf>

- [S8] Skyward secondary guides (district) — e.g. https://www.doversd.org/downloads/documents_and_forms/staff_-_skyward/skyward_gradebook_teacher_guide.pdf
- [S9] Kelyra [research/05-gradebooks-skill-plans-incentives.md](#) (2026-08-12; skimmed 2026-09-24)
- [S10] Kelyra AVG-R1 / AVG-R2 / AVG-P1 company notes (2026-09-02)
- [S11] Kelyra [docs/mvp.md](#), [docs/data-model.md](#), [docs/vision.md](#), [notes/authoring/kinds-metrics.md](#)

All URLs accessed 2026-09-24 CT.

Plan (full SoT)

Gradeable Work Types Plan (GWT-P1)

Date: 2026-09-24 Author: Chief of Staff / Grok Bot (Kelyra) Status: Ready for Chuck review Revision: R1 Cards: [t_a2d9bdc1](#) Research: [notes/company/gradeable-work-types-research.md](#) Stack: Expo + Supabase. No app/SQL from this card.

Product law (short)

Approve gate · not of record · hybrid work_kind + categories · Author stays [kind=lesson](#) · capture stays thin · weights belong to AVG.

Phased delivery

Phase	Scope
A — Labels	work_kind seed, picker, defaults, capture type hint, Christian preset, complete/incomplete
B — Syllabus math	ClassSyllabus + weights + drop/retake (AVG)
C — Marks & elementary	Mark codes, ESNU/letter, checklists
D — SBG / sync	Standards attempts, LMS/SIS

Needed P0

G0-1 work_kind seed + picker · G0-2 default category map · G0-3 capture type hint · G0-4 Approve gate · G0-5 Christian preset · G0-6 honest AVG HOLD · G0-7 no quiz player · G0-8 complete/incomplete · G0-9 participation/behavior default out of average.

Coverage and uncertainty

- Did not independently log into PowerSchool/FACTS tenants; patterns from public docs + school PDFs.
- Skyward admin-lock vs teacher-edit varies by district configuration.
- Spring Baptist exact handbook not in-repo; Christian seeds are informed estimates from FACTS training examples.
- Live DB may differ from [docs/data-model.md](#) — Architect verifies before Eng; this card invents no migrations.

Meta · Card [t_a2d9bdc1](#) (GRADE) · Dark visual edition 2026-09-24 · Not legal advice where privacy/compliance topics appear · No Hermes staffing.